Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption
One potential explanation for the website's connection timeout error message is that the web server is overwhelmed by numerous TCP SYN requests resulting in the web server being overwhelmed by multiple requests and shutting the services down. The logs show that the numerous TCP SYN requests coming from an unfamiliar IP Address can indicate an intrusion attack by a malicious actor. This event could impact the overall services of the company by stagnating the operations that can cause millions of damages due to unavailability to perform the usual routines.

Section 2: Explain how the attack is causing the website to malfunction
When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake: 1. The SYN is the request of the source IP address to the destination web server to verify if the connection is stable. 2. The SYN ACK is the signal from the web server page to say that the request is acknowledged. 3. The ACK is the final form of connection for the source IP Address to access stable connection to the web server. When a malicious actor sends a multiple or numerous request of SYN packets to the server, the server then will be flooded and overwhelmed to read each out, resulting in halt of services. The log shows that the web server was attacked with multiple SYN request from an unfamiliar IP Address that can indicate a malicious attack to the web server.